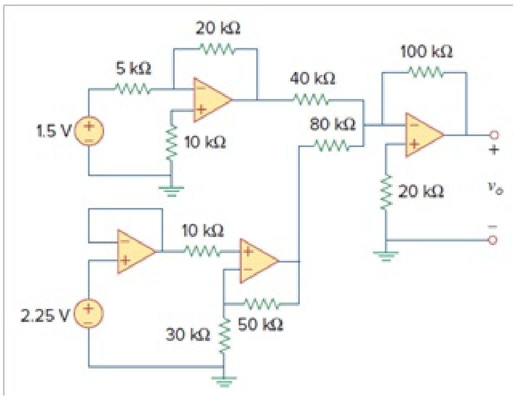
Chapter 5, Problem 71P

Problem

Determine *vo* in the op amp circuit of Fig. 5.97.

Figure. 5.97:



Step-by-step solution

Step 1 of 5

Draw the circuit diagram as shown in Figure 1.

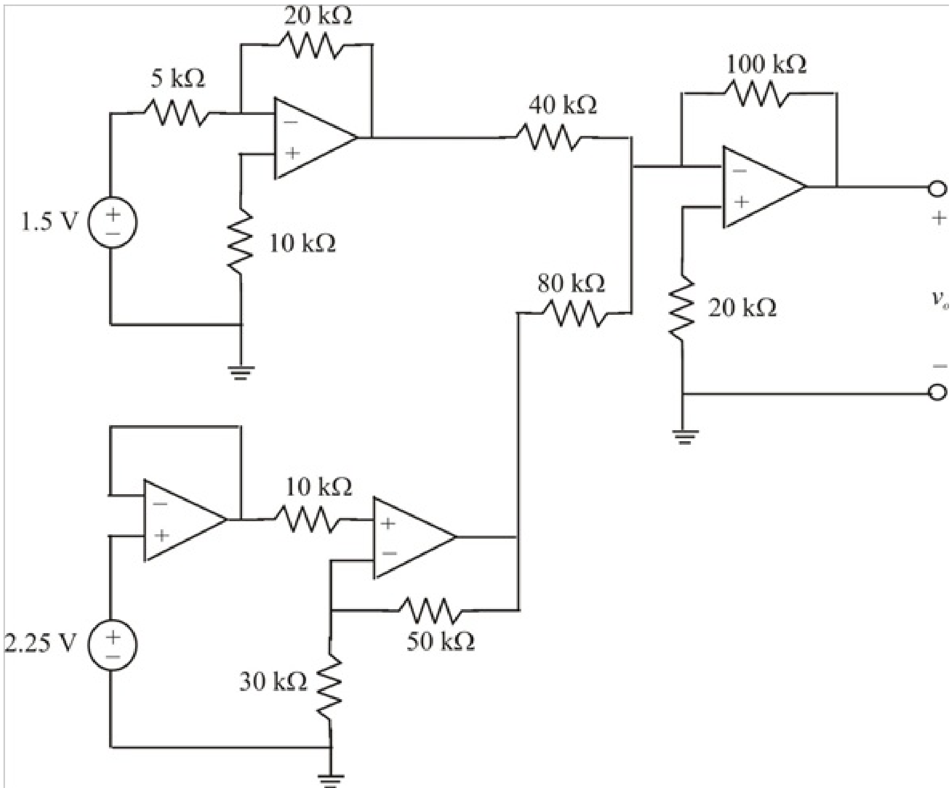


Figure 1

Step 2 of 5

The circuit is labeled as shown in Figure 2.

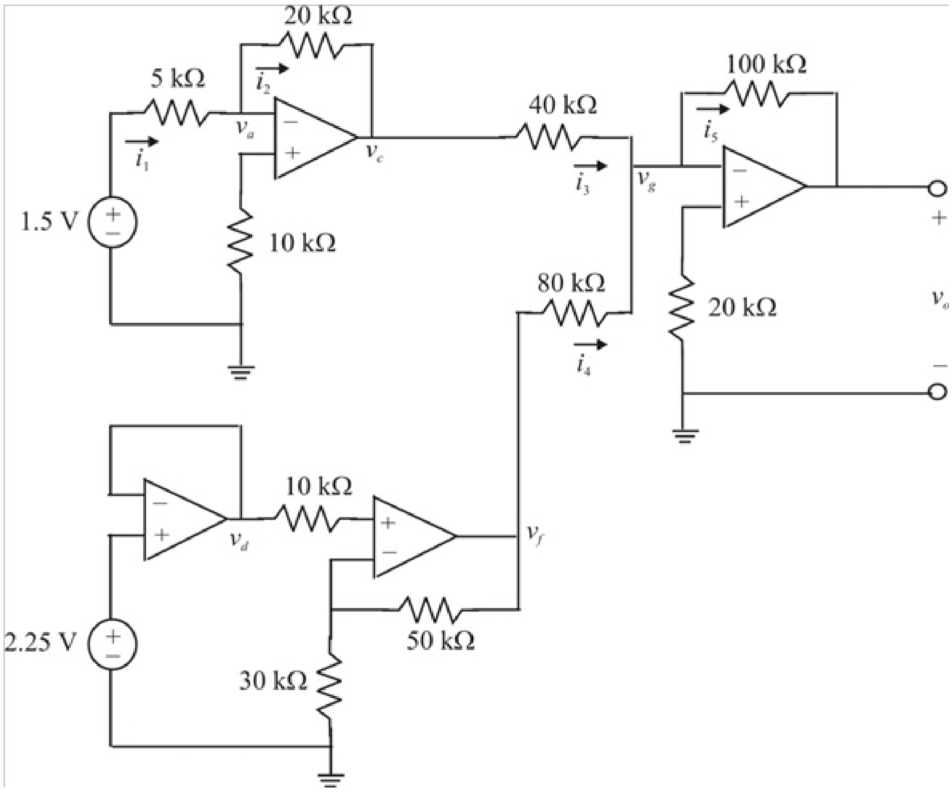


Figure 2

Step 3 of 5

Applying Kirchhoff's current law to node v_a,

$$i_1 = i_2$$
$$\frac{1.5 - v_a}{5} = \frac{v_a - v_c}{20}$$
$$6 - 4v_a = v_a - v_c$$
$$5v_a - v_c = 6$$

Therefore $5v_a - v_c = 6$ …… (1)

But $v_a = 0$ since non-inverting terminal is grounded. Then equation (1) becomes,

$$5v_a - v_c = 6$$
$$5(0) - v_c = 6$$
$$v_c = -6\,\text{V}$$

Therefore $v_c = -6\,\text{V}$ …… (2)

Step 4 of 5

For the voltage follower, the voltage v_d is:

$$v_d = 2.25\,\text{V}$$

Therefore $v_d = 2.25\,\text{V}$ …… (3)

The output voltage v_f of a non-inverting amplifier is:

$$v_f = \left(1 + \frac{R_f}{R_1}\right) v_i$$

Where

$$R_f = 50\,\text{k}\Omega$$
$$R_1 = 30\,\text{k}\Omega$$
$$v_i = v_d = 2.25\,\text{V}$$

The voltage v_f is:

$$\begin{aligned} v_f &= \left(1 + \frac{50}{30}\right)(2.25) \\ &= 6\,\text{V} \end{aligned}$$

Therefore $v_f = 6\,\text{V}$ …… (4)

Applying Kirchhoff's current law to node v_g,

$$i_3 + i_4 = i_5$$
$$\frac{v_c - v_g}{40} + \frac{v_f - v_g}{80} = \frac{v_g - v_o}{100}$$
$$10v_c - 10v_g + 5v_f - 5v_g = 4v_g - 4v_o$$
$$10v_c - 19v_g + 5v_f = -4v_o$$

Therefore $10v_c - 19v_g + 5v_f = -4v_o$ …… (5)

Step 5 of 5

But $v_g = 0$ since non-inverting terminal is grounded. Then equation (5) becomes,

$$10v_c - 19v_g + 5v_f = -4v_o$$
$$10v_c - 19(0) + 5v_f = -4v_o$$
$$10v_c + 5v_f = -4v_o$$

Therefore $10v_c + 5v_f = -4v_o$ …… (6)

Substituting equation (2) and (4) in equation (6), we get

$$10v_c + 5v_f = -4v_o$$
$$10(-6) + 5(6) = -4v_o$$
$$-60 + 30 = -4v_o$$
$$v_o = 7.5\,\text{V}$$

Therefore the output voltage v_o is $\boxed{7.5\,\text{V}}$.